Algebra 2 Review: Quadratic Functions & Complex Numbers

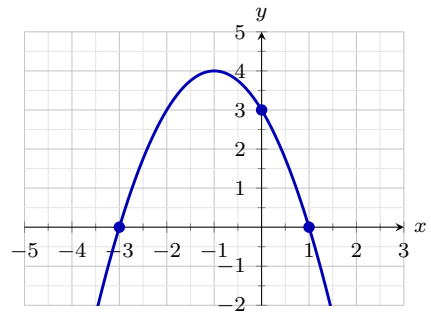
Station 1 of 4: Quadratic Functions, Forms, and Graphs

Name: _____ Date: _____

Match each form to what it shows: vertex form gives the vertex, factored form gives the intercepts. Remember $(x-h)$ means the vertex is at $x=h$.

1. For $f(x) = 2x^2 - 8x + 3$, write f in vertex form, then state the vertex, the axis of symmetry, and the minimum value.

2. The parabola shown crosses the axes at the marked points. Write its equation in factored form and in standard form, then find the vertex.



3. Describe, in order, the transformations of $y = x^2$ that give $g(x) = -\frac{1}{2}(x+3)^2 - 2$, and state the vertex and whether the parabola opens up or down.

4. Factor $f(x) = 2x^2 - 4x - 30$ completely, then find the x-intercepts, the axis of symmetry, and the vertex.

5. A parabola has x-intercepts -2 and 5 and passes through $(1, -24)$. Write its equation in factored form and in standard form.

6. A parabola has vertex $(-3, 2)$ and passes through $(-1, -6)$. Write its equation in vertex form.

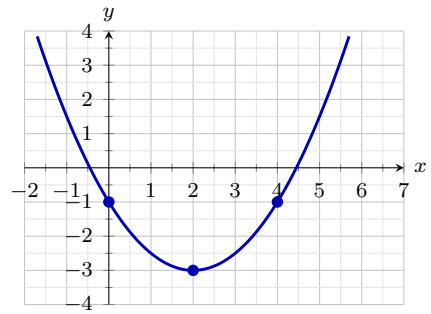
7. Let $f(x) = x^2 - 10x + 30$. The table shows a quadratic function g. Which function has the greater minimum value? Give each minimum and where it occurs.

x	0	1	2	3	4
$g(x)$	9	3	1	3	9

8. Find the range of $f(x) = -2x^2 + 12x - 7$ (a) for all real x and (b) on the restricted domain $0 \le x \le 2$.

9. Find the quadratic $y = ax^2 + bx + c$ whose graph passes through $(-1, 6)$, $(0, 1)$, and $(2, 3)$.

10. The graph shows a parabola with vertex $(2, -3)$ and the points $(0, -1)$ and $(4, -1)$. Write its equation in vertex form and in standard form.



Algebra 2 Review: Quadratic Functions & Complex Numbers

Station 2 of 4: Solving Quadratic Equations and the Discriminant

Name: _____ Date: _____

Move everything to one side before factoring. Keep the ±, divide the whole numerator, and check for extraneous solutions.

1. Solve $6x^2 - 5x - 4 = 0$ by factoring.

2. Solve $2(x+1)^2 = 36$ by taking square roots. Give exact answers in simplest radical form.

3. Solve $x^2 + 10x - 3 = 0$ by completing the square.

4. Solve $2x^2 - 12x + 7 = 0$ by completing the square. (Factor out the 2 first.)

5. Use the quadratic formula to solve $2x^2 + 3x - 7 = 0$.

6. Find the discriminant of each equation and describe its solutions. (a) $x^2 - 6x + 9 = 0$ (b) $2x^2 + x + 5 = 0$ (c) $3x^2 - x - 4 = 0$.

7. For $x^2 - 4x + k = 0$, find the values of k that give (a) two distinct real solutions, (b) exactly one real solution, (c) two nonreal solutions.

8. Solve $x^4 - 13x^2 + 36 = 0$. (Let $u = x^2$.)

9. Solve $(x+2)(x-3) = 14$. Explain why you cannot set each factor equal to 14.

10. Solve $\sqrt{x+6} = x$ and check for extraneous solutions.

Algebra 2 Review: Quadratic Functions & Complex Numbers

Station 3 of 4: Complex Numbers

Name: _____ Date: _____

Use $i^2 = -1$ and the powers of i repeat every 4. To divide, multiply the top and bottom by the conjugate of the denominator.

1. Simplify (a) $\sqrt{-48}$ (b) $\sqrt{-9} \cdot \sqrt{-4}$. Why is (b) not $\sqrt{36}$?

2. Simplify (a) i^{37} (b) i^{102} (c) $\dfrac{1}{i^3}$.

3. Simplify $(7 - 3i) - (2 - 8i) + (-4 + i)$.

4. Multiply $(2 + 5i)(3 - 4i)$ and write the result as $a + bi$.

5. Simplify $(4 - 3i)^2$.

6. Write the conjugate of $-2 - 7i$ and find the product of the number and its conjugate.

7. Write $\dfrac{4 + 7i}{3 - 2i}$ in the form $a + bi$.

8. Solve $x^2 - 4x + 9 = 0$ over the complex numbers.

9. Solve $3x^2 + 2x + 5 = 0$. Write each solution as $a + bi$.

10. A quadratic with real coefficients has the root $3 - 4i$. State the other root, then write the monic quadratic in standard form.

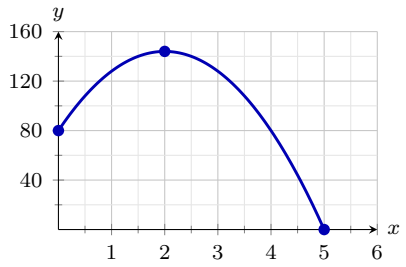
Algebra 2 Review: Quadratic Functions & Complex Numbers

Station 4 of 4: Quadratic Models, Inequalities, and Systems

Name: _____ Date: _____

Interpret the vertex in context, reject impossible values such as negative time, and test a point to choose the region of an inequality.

1. A ball is thrown upward from a platform. Its height is $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find the maximum height, when it occurs, and when the ball lands.



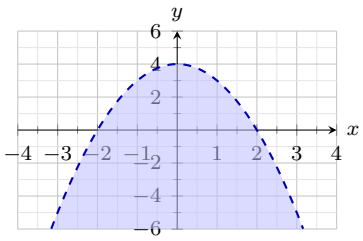
2. A theater sells $500 - 20p$ tickets when the price is p dollars. Write the revenue $R(p)$ and find the price that maximizes it and the maximum revenue.

3. A rancher has 240 ft of fence for a rectangular field divided into two pens by one fence parallel to the width. Find the dimensions and the largest area.

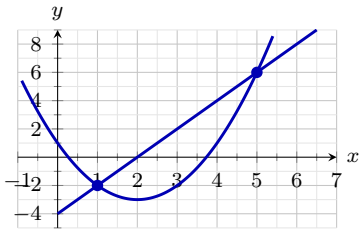
4. A company's profit is $P(x) = -2x^2 + 28x - 66$ thousand dollars when x hundred items are sold. For which values of x is the profit positive?

5. Solve each inequality and write the solution in interval notation. (a) $x^2 - 3x - 10 \geq 0$ (b) $-x^2 + 4x + 5 > 0$.

6. The shaded region (dashed boundary) is the graph of a quadratic inequality. Write the inequality. Is $(0, 0)$ a solution? Is $(3, 0)$?



7. The graph shows $y = x^2 - 4x + 1$ and $y = 2x - 4$. Solve the system algebraically and confirm that your solutions match the intersections.



8. How many solutions does $y = x^2 - 4x + 5$ have with each line? (a) $y = 2x - 4$ (b) $y = 2x - 10$ (c) $y = 2x$. Find any solutions.

9. A sprinkler's water is at height y feet when it is x feet away horizontally. Use quadratic regression, rounding coefficients to the nearest tenth, then find the maximum height and where it occurs (nearest tenth).

x	0	1	2	3	4	5
y	5	13	18	19	16	9

10. An arch has its vertex at $(0, 30)$ and meets the ground at $x = \pm 20$ (feet). Write its equation, find its height 10 ft from the center, and find its width at a height of 15 ft (nearest tenth).

Algebra 2 Review: Quadratic Functions & Complex Numbers

Answer Key

Station 1: Quadratic Functions, Forms, and Graphs

1. $f(x) = 2(x-2)^2 - 5$; vertex $(2, -5)$; axis $x = 2$; minimum value -5
   $2(x^2 - 4x + 4) - 8 + 3$
2. $y = -(x+3)(x-1) = -x^2 - 2x + 3$; vertex $(-1, 4)$
   $a(0+3)(0-1) = 3 \Rightarrow a = -1$; *axis is* $x = \dfrac{-3+1}{2}$
3. reflect over the x-axis, vertical compression by $\frac{1}{2}$, left 3, down 2; vertex $(-3, -2)$; opens down
   $x + 3$ *shifts left, not right*
4. $f(x) = 2(x-5)(x+3)$; intercepts 5 and -3; axis $x = 1$; vertex $(1, -32)$
   axis is halfway between the intercepts
5. $y = 2(x+2)(x-5) = 2x^2 - 6x - 20$
   $-24 = a(3)(-4) \Rightarrow a = 2$
6. $y = -2(x+3)^2 + 2$
   $-6 = a(2)^2 + 2 \Rightarrow a = -2$
7. f: minimum 5 at $x = 5$; g: minimum 1 at $x = 2$; f has the greater minimum
   the table is symmetric about $x = 2$
8. (a) $y \le 11$; (b) $-7 \le y \le 9$ (still increasing on $[0, 2]$, so use the endpoints)
   vertex is $(3, 11)$, *outside* $[0, 2]$
9. $y = 2x^2 - 3x + 1$
   $c = 1$; *then* $a - b = 5$ *and* $2a + b = 1$
10. $y = \dfrac{1}{2}(x-2)^2 - 3 = \dfrac{1}{2}x^2 - 2x - 1$
    $-1 = a(0-2)^2 - 3 \Rightarrow a = \dfrac{1}{2}$

Station 2: Solving Quadratic Equations and the Discriminant

1. $x = \dfrac{4}{3}$ or $x = -\dfrac{1}{2}$
   $(3x - 4)(2x + 1) = 0$
2. $x = -1 \pm 3\sqrt{2}$
   $(x+1)^2 = 18$ *and* $\sqrt{18} = 3\sqrt{2}$
3. $x = -5 \pm 2\sqrt{7}$
   $x^2 + 10x + 25 = 3 + 25$; $(x+5)^2 = 28$
4. $x = 3 \pm \dfrac{\sqrt{22}}{2}$
   $(x-3)^2 = \dfrac{11}{2}$
5. $x = \dfrac{-3 \pm \sqrt{65}}{4}$
   $b^2 - 4ac = 9 + 56 = 65$
6. (a) 0: one repeated real solution; (b) -39: two nonreal complex solutions; (c) 49: two rational real solutions
   in (c), $b = -1$ *so* $b^2 = 1$
7. $b^2 - 4ac = 16 - 4k$: (a) $k < 4$; (b) $k = 4$; (c) $k > 4$
8. $x = \pm 2$ or $x = \pm 3$
   $(u-4)(u-9) = 0$
9. $x = 5$ or $x = -4$
   move everything to one side first: $x^2 - x - 20 = 0$
10. $x = 3$ (since $x = -2$ is extraneous: $\sqrt{4} = 2 \neq -2$)
    squaring gives $x^2 - x - 6 = 0$

Station 3: Complex Numbers

1. (a) $4i\sqrt{3}$; (b) $(3i)(2i) = 6i^2 = -6$
   write each root with i *before multiplying*
2. (a) i; (b) -1; (c) i
   divide the exponent by 4 and use the remainder; $\dfrac{1}{i^3} = \dfrac{1}{-i} = i$
3. $1 + 6i$
   distribute the minus sign to both parts of $2 - 8i$
4. $26 + 7i$
   $6 - 8i + 15i - 20i^2$
5. $7 - 24i$
   $16 - 24i + 9i^2$; *do not square each term only*
6. conjugate $-2 + 7i$; product 53
   $(-2)^2 + 7^2$
7. $-\dfrac{2}{13} + \dfrac{29}{13}i$
   multiply by $\dfrac{3+2i}{3+2i}$; *denominator* $9 + 4 = 13$
8. $x = 2 \pm i\sqrt{5}$
   $\dfrac{4 \pm \sqrt{-20}}{2}$
9. $x = \dfrac{-1 \pm i\sqrt{14}}{3}$
   $\dfrac{-2 \pm \sqrt{-56}}{6}$ *and* $\sqrt{-56} = 2i\sqrt{14}$
10. other root $3 + 4i$; $x^2 - 6x + 25 = 0$
    sum 6, *product* $9 + 16 = 25$

Station 4: Quadratic Models, Inequalities, and Systems

1. maximum 144 ft at $t = 2$ s; lands at $t = 5$ s (reject $t = -1$)
   $-16(t-5)(t+1) = 0$
2. $R(p) = 500p - 20p^2$; price \$12.50; maximum revenue \$3,125
   vertex at $p = \dfrac{-500}{2(-20)}$
3. 40 ft by 60 ft; maximum area 2,400 ft^2
   $3x + 2L = 240$, *so* $A = x \cdot \dfrac{240 - 3x}{2}$
4. $3 < x < 11$: between 300 and 1,100 items
   $P = -2(x-3)(x-11)$
5. (a) $(-\infty, -2] \cup [5, \infty)$; (b) $(-1, 5)$
   in (b), multiply by -1 *and reverse the inequality*
6. $y < -x^2 + 4$; $(0,0)$: $0 < 4$ is true, so yes; $(3,0)$: $0 < -5$ is false, so no
   dashed boundary means strict inequality
7. $(1, -2)$ and $(5, 6)$
   $x^2 - 6x + 5 = 0$, *so* $(x-1)(x-5) = 0$
8. (a) one: $(3, 2)$; (b) none; (c) two: $(1, 2)$ and $(5, 10)$
   discriminants of $x^2 - 6x + c$: 0, -24, 16
9. $y = -1.9x^2 + 10.4x + 4.8$; maximum ≈ 19.0 ft at $x \approx 2.7$ ft
   $x = \dfrac{-10.4}{2(-1.9)}$
10. $y = -\dfrac{3}{40}x^2 + 30$; height 22.5 ft; width $20\sqrt{2} \approx 28.3$ ft
    $0 = a(400) + 30 \Rightarrow a = -\dfrac{3}{40}$